

SE 410 - Software Framework Applications

Requirements Document

Sustainable Product Shopping Platform with Carbon Tracking

1 Scenario 1: Corporate Sustainable Procurement by Delta Air Lines

Entity: Delta Air Lines – Procurement and Sustainability Department

Goal: To procure bulk quantities of eco-friendly cabin products while accurately calculating the total carbon footprint and generating reports for the airline’s sustainability initiatives.

Context: Delta Air Lines is conducting its quarterly bulk purchasing for cabin materials. The procurement and sustainability teams access the platform through a corporate account using company laptops. They expect fast response times, secure transactions, data accuracy, and an intuitive interface.

Narrative:

Delta Air Lines logs into the platform using its dedicated corporate account with proper role-based access. The team browses the Product Catalog, where all pages load within 2 seconds. They search and select 1,200 units of lightweight recyclable seat covers, 3,000 sets of sustainable crew uniforms, and 25,000 biodegradable meal trays. Each product clearly displays name, description, price, stock quantity, and carbon footprint value per unit.

As the team adds items to the corporate cart, the Carbon Footprint Calculator instantly updates the total carbon footprint of the order in real time. The system automatically validates stock levels and prevents any out-of-stock purchases.

The team then proceeds to checkout, reviews the complete order summary (items, quantities, total price, and total carbon impact), and completes the secure payment. The transaction is processed successfully with full encryption and rollback protection. Immediately after, automated email and in-app notifications are sent to the relevant team members.

Over the next few days, the team monitors the order through the Order Tracking module. After delivery, they access the Carbon Footprint Dashboard to view the detailed impact of this order, monthly aggregation, and cumulative carbon footprint. They export accurate reports for ESG documentation. All carbon calculations are consistent and the interface remains user-friendly across devices.

Outcome: Delta Air Lines successfully completes a high-volume sustainable procurement efficiently and securely. The platform delivers strong performance, reliable carbon tracking, and excellent usability, supporting the airline’s environmental goals and reporting needs.